```
#include <SPI.h>  // Include the RFID libraries
#include <MFRC522.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <stdio.h>
#include <string.h>
#define SS_PIN 10    // Declare the reset and SDA pins of RFID
#define RST_PIN 9
MFRC522 mfrc522(SS_PIN, RST_PIN);  // Create MFRC522 instance
LiquidCrystal_I2C lcd(0x27, 16, 2);  // (I2C addr., rows, columns)
```

```
byte validUIDs[][4] = {
  {0xA3, 0xA8, 0x7F, 0xFA}, // UID of student 1 TAG
  {0x9C, 0xD0, 0xB8, 0x03}, // UID of student 2 TAG
  {0xAC, 0x99, 0xE1, 0x03}, // UID of student 3 TAG
};
const char* passNames[] = {
  "CHIBUZE",   // Name of student 1
  "MIMO",      // Name of student 2
  "CHAKI",     // Name of student 3
};
```

```
String currPass[sizeof(validUIDs)];
// Array to store currently
authorized UIDs
int bal[] = {100, 100, 50};
int dist[] = {1, 1, 1};
int rate = 10;
int total_dist = 0;


void setup() {
  Serial.begin(9600);  // Initiate SPI
bus
  SPI.begin();
  mfrc522.PCD_Init();  // Initiate
MFRC522
```

```
  lcd.begin(16, 4);    // Begin the
LCD
  lcd.init();
  lcd.backlight();
  pinMode(A0, OUTPUT); //
Actuator buzzer pin


  lcd.clear();
  lcd.setCursor(5, 0);
  lcd.print("WELCOME");
  lcd.setCursor(7, 1);
  lcd.print("TO");
  delay(2500);
  lcd.setCursor(0, 0);
```

```
  lcd.print("   BUS TICKET   ");
  lcd.setCursor(0, 1);
  lcd.print(" PAYMENT SYSTEM ");
  delay(2500);
  lcd.clear();
  lcd.setCursor(2, 0);
  lcd.print("Bus Ticketing");
  Serial.print("Ready to read RFID
cards");
  delay(2000);
  lcd.setCursor(0, 0);
  lcd.print("Bus Fare:$10    ");
  lcd.print(rate);
  lcd.setCursor(0, 1);
```

```
  lcd.print("Dist. Covered:");
  lcd.print(total_dist);
  delay(2000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("  SCAN TAG FOR");
  lcd.setCursor(0, 1);
  lcd.print("   BUS TICKET");
  delay(1000);
}
void checkPass(String content) {
  bool foundInBus = false;
  for (int i = 0; i < sizeof(currPass);
i++) {
```

```
  if (currPass[i] == content) { // Check if UID is already authorized (scanned once)
    foundInBus = true;
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("ThankYou ");
    lcd.print(passNames[i]);
    lcd.setCursor(0, 1);
    lcd.print("Balance:$");
    bal[i] = bal[i] - dist[i]*rate;
    lcd.print(bal[i]); // Display remaining balance
```

```
    currPass[i] = ""; // Remove the authorized UID from currPass (for next scan)
    tone(A0, 1000);
    delay(1000);
    noTone(A0);
    delay(2000);
    break;
  }
  else if(currPass[i+1] == content){
    foundInBus = true;
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("ThankYou ");
```

```
    lcd.print(passNames[i+1]);
    lcd.setCursor(0, 1);
    lcd.print("Balance:$");
    bal[i+1] = bal[i+1] -
dist[i+1]*rate;
    lcd.print(bal[i+1]); // Display
remaining balance
    currPass[i+1] = ""; // Remove
the authorized UID from currPass
(for next scan)
    tone(A0, 1000);
    delay(1000);
    noTone(A0);
    delay(2000);
```

```
      break;
    }
    else if(currPass[i+2] == content){
      foundInBus = true;
      lcd.clear();
      lcd.setCursor(0, 0);
      lcd.print("ThankYou ");
      lcd.print(passNames[i+2]);
      lcd.setCursor(0, 1);
      lcd.print("Balance:$");
      bal[i+2] = bal[i+2] -
dist[i+2]*rate;
      lcd.print(bal[i+2]); // Display
remaining balance
```

```
    currPass[i+2] = ""; // Remove
the authorized UID from currPass
(for next scan)
    tone(A0, 1000);
    delay(1000);
    noTone(A0);
    delay(2000);
    break;
  }
  if (!foundInBus){
    for (int i = 0; i <
sizeof(validUIDs); i++) {
      if
(memcmp(mfrc522.uid.uidByte,
```

```
validUIDs[i], mfrc522.uid.size) ==
0){
          Serial.print(currPass[i]); //
Add the UID to currPass (for
subsequent scans)
          lcd.clear();
          lcd.setCursor(0, 0);
          lcd.print("Welcome
Aboard");
          lcd.setCursor(0, 1);
          lcd.print(passNames[i]);
          tone(A0, 1000);
          delay(500);
          noTone(A0); // Welcome
message with student name
```

```
          delay(1000);
          currPass[i] = content;
          break;
        }
        else{
          continue;
        }
        break;
    }
   }
    break;
  }
}
```

```
void unregistered() {
  lcd.clear();
  for (int i = 0; i < 3; i++) {
    tone(A0, 1000);
    lcd.print("  UNREGISTERED!");
    delay(500);
    noTone(A0);
    lcd.clear();
    delay(500);
  }
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(" UNREGISTERED!");
```

```
  lcd.setCursor(0, 1);
  lcd.print("PLS GET A VALID
CARD");
  delay(700);
  for (int posCounter = 0;
posCounter < 43; posCounter++) {
    lcd.scrollDisplayLeft(); // Scroll
one position left
    delay(200);             // Wait a bit
  }
}

void loop() {
  SPI.begin();       // Initiate SPI bus
```

```
  mfrc522.PCD_Init();  // Initiate
MFRC522
  // Look for new cards
  if
(mfrc522.PICC_IsNewCardPresent(
)){
    if
(mfrc522.PICC_ReadCardSerial()) {
      Serial.print("\nUID tag: \n");
      for (byte i = 0; i <
mfrc522.uid.size; i++) {
        mfrc522.uid.uidByte[i] =
toupper(mfrc522.uid.uidByte[i]);
        Serial.print("0x");
```

```
      if (mfrc522.uid.uidByte[i] <
0x10) Serial.print("0");

Serial.print(mfrc522.uid.uidByte[i]
, HEX);
      if (i < mfrc522.uid.size - 1)
Serial.print(", ");
    }


    String content = "";
    for (byte i = 0; i <
mfrc522.uid.size; i++) {
      content +=
String(mfrc522.uid.uidByte[i] <
0x10 ? "0" : " ") +
```

```
String(mfrc522.uid.uidByte[i],
HEX);
    }

    bool validUID = false;
    bool foundInBus = false;
    for (int i = 0; i <
sizeof(validUIDs); i++) {
      if
(memcmp(mfrc522.uid.uidByte,
validUIDs[i], mfrc522.uid.size) ==
0) {
        validUID = true;
        break;
      }
```

```
}
checkPass(content);
if (!validUID){
  unregistered();
}


lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Bus Fare:$10    ");
lcd.print(rate);
lcd.setCursor(0, 1);
lcd.print("Dist. Covered:");
total_dist += 10;
```

```
  for (int i=0; i< sizeof(dist); i++){
    dist[i] += 10;
  }
  lcd.print(total_dist);
  delay(2000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("  SCAN TAG FOR");
  lcd.setCursor(0, 1);
  lcd.print("   BUS TICKET");
  delay(1000);


}
```

```
    mfrc522.PICC_HaltA();   // Stop
reading
    mfrc522.PCD_StopCrypto1();
    }
}
```